

1. employees above their department average

```
select emp_id, emp_name, dept_id, salary from employees e where salary > (select avg(salary) from employees where dept_id = e.dept_id);
```

2. highest paid employee(s) in every department

```
select emp_id, emp_name, dept_id, salary from employees e where salary = (select max(salary) from employees where dept_id = e.dept_id);
```

3. second highest distinct salary

```
select max(salary) as second_highest_salary from employees where salary < (select max(salary) from employees);
```

4. employees receiving the second highest distinct salary

```
select emp_name, salary from employees where salary = (select max(salary) from employees where salary < (select max(salary) from employees));
```

5. employees earning more than their manager

```
select emp_id, emp_name, salary, manager_id from employees e where manager_id is not null and salary > (select salary from employees m where m.emp_id = e.manager_id);
```

6. managers who earn less than at least one direct report

```
select emp_id, emp_name, salary from employees e where exists (select 1 from employees sub where sub.manager_id = e.emp_id and sub.salary > e.salary);
```

7. departments whose average salary is above company average

```
select dept_id from employees group by dept_id having avg(salary) > (select avg(salary) from employees);
```

8. employees in departments whose average salary is above company average

```
select emp_id, emp_name, dept_id, salary from employees where dept_id in (select dept_id from employees group by dept_id having avg(salary) > (select avg(salary) from employees));
```

9. employees earning more than every employee in hr

```
select emp_id, emp_name, salary from employees where salary > all (select salary from employees where dept_id = (select dept_id from departments where dept_name = 'hr'));
```

10. employees earning more than at least one research employee

```
select emp_id, emp_name, salary from employees where salary > any (select salary from
employees where dept_id = (select dept_id from departments where dept_name =
'research'));
```

11. departments having no projects

```
select dept_id, dept_name from departments d where not exists (select 1 from projects p
where p.dept_id = d.dept_id);
```

12. employees belonging to departments with more than one project

```
select emp_id, emp_name, dept_id from employees e where (select count(*) from projects p
where p.dept_id = e.dept_id) > 1;
```

13. employees in the department of the highest-budget project

```
select emp_id, emp_name, dept_id, salary from employees where dept_id in (select dept_id
from projects where budget = (select max(budget) from projects));
```